

ronuku@umich.edu

RAPHAEL S. ONUKU

Ann Arbor, MI • www.linkedin.com/in/raphaelonuku

CAREER OBJECTIVE

• To contribute effectively to the scientific research and strategic planning objectives of any institution I come in contact with and strive for excellence and precision at all times, in all positions and circumstances, attaining professional distinction and proficiency.

EDUCATION

- 1. University of Michigan, Ann Arbor, USA** **2023 - Date**
PhD Candidate, Medicinal Chemistry,
M.S. in Medicinal Chemistry GPA 3.62/4.00
- 2. Taipei Medical University, Taipei, Taiwan** **2021 - 2023**
Master of Science in Pharmacy (Medicinal Chemistry), GPA 4.00/4.00
Thesis: [Rational Design of Celecoxib-Based HDAC Inhibitors as Potent Anti-Glioblastoma Agents.](#)
- 3. University of Nigeria, Nsukka, Nigeria** **2011 - 2017**
Bachelor of Pharmacy Degree (BPharm)
Thesis: Repurposing Drugs to Find HIV-1 Protease Inhibitors: A Virtual Study.

RESEARCH/LABORATORY EXPERIENCE

- 1. Graduate Student Research Assistant**, Department of Medicinal Chemistry, University of Michigan, Ann Arbor, USA
• Cichewicz Lab, **June 2024 - Current**
Projects: *Peptide synthesis and applied computational drug discovery research.*
- 2. Graduate Research Rotations**, University of Michigan, Ann Arbor, USA
• Brooks Lab, **Jan 2024 - Apr 2024**
Project: *Comparative Evaluation of CDocker and AutoDock Vina Docking Protocols Using Tanimoto Coefficient-Based Selection of Active and Inactive Compounds for PIN1 Target.*

• Cernak Lab, **Oct 2023 - Dec 2023**
Project: *From Sequence to Structure: Structural Prediction and Analysis of CTHRC1 in Sea Turtle Fibropapillomatosis Using Homology Modeling.*

• Sztain Lab, **Aug 2023 - Oct 2023**
Project: *Investigating the Binding and Interaction of Nirmatrevir To SARS-Cov-2 Main Protease (M^{pro}) Using Molecular Dynamics Approach.*
- 3. Pre-Doctoral Intern**, Institute of Chemistry, Academia Sinica, Taiwan
• David Wu's Lab, **Jun 2023 - July 2023.**
Project: *Exploring Hydrogel Formation of A₃K₃ Peptide Through Molecular Dynamics Simulation.*
- 4. Graduate Research Assistant**, Research Center of Pharmaceutical Sciences and Biotechnology, Taipei Medical University, Taipei, Taiwan, **2021 - 2023.**

5. **Research Assistant**, Department of Pharmaceutical and Medicinal Chemistry, Faculty of Pharmaceutical Chemistry, University of Nigeria, **2019 - 2024**.

6. **Voluntary Research Assistant/Intern Pharmacist**, Department of Pharmaceutical and Medicinal Chemistry, Faculty of Pharmaceutical Chemistry, University of Nigeria, **2018 - 2019**.

7. **Student Research Assistant**, Department of Pharmaceutical and Medicinal Chemistry, Faculty of Pharmaceutical Chemistry, University of Nigeria, **2016 - 2017**.

TEACHING EXPERIENCE

Pharmacy Preceptor (NYSC), Dept. of Pharmacy, Univ. of Uyo Teaching Hospital, Nigeria, **2020 - 2021**.

- Mentored intern pharmacists through small-group tutorials and coaching.

Academic Tutor, Faculty of Pharmacy, Univ. of Nigeria, Nsukka, **2017 - 2021**.

- Prepared materials and ran peer tutorials for prospective and current undergraduates.

Volunteer STEM Instructor, Model Secondary School, Nsukka, **2012 - 2016**.

- Led senior-level tutorials and organized student study groups.

PUBLICATIONS

Peer-reviewed journal articles

1. Shen CJ, Chen HC, Lin CL, Thakur A, **Onuku R**, Chen IC, Li HY, Chang KY, Chuang JY, Yang WB, Lin HY, Shen YR, Liou JP, Chang WC, Hsu TI. Contribution of Prostaglandin E2-Induced Neuronal Excitation to Drug Resistance in Glioblastoma Countered by a Novel Blood-Brain Barrier Crossing Celecoxib Derivative. *Adv Sci* (Weinh). 2025 Jul 14:e06336. doi:[10.1002/advs.202506336](https://doi.org/10.1002/advs.202506336).

2. Mishra A, Thakur A, Sharma R, **Onuku RS**, Kaur C, Liou JP, Hsu SP, Nepali K. Scaffold hopping approaches for dual-target antitumor drug discovery: opportunities and challenges. *Expert Opin Drug Discov*. 2024 Nov;19(11):1355-1381. doi:[10.1080/17460441.2024.2409674](https://doi.org/10.1080/17460441.2024.2409674).

3. Ibezim A, **Onuku RS**, Ottih C, Ezeonu I, Onoabedje E, Ramanathan K, Nwodo N. New sulphonamide-peptide hybrid molecules as potential PBP2a ligands and methicillin-resistant Staphylococcus aureus actives. *J Biomol Struct Dyn*. 2023 Aug-Sep;41(14):6684-6694. doi:[10.1080/07391102.2022.2111359](https://doi.org/10.1080/07391102.2022.2111359).

4. Nnadi CO, **Onuku RS**, Ayoka TO, Okorie NH, Nwodo NJ. Chemical constituents of Combretum dolichopetalum: characterization, antitrypanosomal activities and molecular docking studies. *Trop J Pharm Res*. 2022;21(4):801-808. doi:[10.4314/tjpr.v21i4.17](https://doi.org/10.4314/tjpr.v21i4.17).

5. Ekpo MD, Xie J, Liu X, **Onuku RS**, Boafu GF, Tan S. Incorporating cryopreservation evaluations into the design of cell-based drug delivery systems: an opinion paper. *Front Immunol*. 2022;13:967731. doi:[10.3389/fimmu.2022.967731](https://doi.org/10.3389/fimmu.2022.967731).

6. Ibezim A, **Onuku RS**, Ibezim A, Ntie-Kang F, Nwodo NJ, Adikwu MU. Structure-based virtual screening and molecular dynamics simulation studies to discover new SARS-CoV-2 main protease inhibitors. *Sci Afr*. 2021 Nov;14:e00970. doi:[10.1016/j.sciaf.2021.e00970](https://doi.org/10.1016/j.sciaf.2021.e00970).

7. Ntie-Kang F, Telukunta KK, Fobofou SAT, Osamor VC, Egieyeh SA, Valli M, et al. Computational Applications in Secondary Metabolite Discovery (CAiSMD): an online workshop. *J Cheminform*. 2021 Sep

6;13(1):64. doi:[10.1186/s13321-021-00546-8](https://doi.org/10.1186/s13321-021-00546-8). Note: author list includes **Onuku RS**.

8. Singla RK, et al. The International Natural Product Sciences Taskforce and the power of Twitter networking exemplified through #INPST hashtag analysis. *Phytomedicine*. 2023;108:154520. doi:[10.1016/j.phymed.2022.154520](https://doi.org/10.1016/j.phymed.2022.154520). Note: author list includes **Onuku RS**.

INVENTION DISCLOSURE

Onuku R. Institutional invention disclosure from master's thesis: "*Celecoxib-Based HDAC Inhibitors as Potent Anti-Glioblastoma Agents*." Taipei Medical University, 2023. Listed on disclosure record; inventors of record: Liou JP; Hsu TI.

HONORS, AWARDS AND GRANTS

Grants

- Rackham Conference Travel Grant, University of Michigan, **2024**.
- Travel and Logistics Grant, Enugu State Scholarship and Education Loans Board (ESSELB), Enugu State Government, Nigeria, **2021**.

Scholarships and Fellowships

- Eugene and Shirley Cordes Scholarship, College of Pharmacy, Department of Medicinal Chemistry, University of Michigan, **2023**.
- Taiwan Government Scholarship, Ministry of Education, Republic of China (Taiwan), **2021**.

Awards

- Outstanding Graduate Student Award, Ministry of Education and Ministry of Foreign Affairs, Republic of China (Taiwan), **2023**.
- Falling Walls Lab Taipei Award Winner, "Breaking the Wall of Multi-Drug Resistant Malaria," **2022**.

Recognitions and Certificates

- Certificate of Achievement, College of Pharmacy, for successful advancement of candidacy, **2025**.
- Certificate of Appreciation, Taiwan Scholarship and Huayu Enrichment Scholarship Program, for contributions to the 2022 Orientation and Welcome Party, **2022**.
- Certificate of Internship, Campus Ambassador Intern, International Model United Nations, **2021**.

CONFERENCES

- 28th International Electronic Conference on Synthetic Organic Chemistry, virtual, **Nov 15 to 30, 2024**.
- Computational Medicinal Chemistry School, Novartis Institutes for Biomedical Research, Cambridge MA, **Oct 28 to 30, 2024**.
- 16th EFMC-YSN MedChemBioOnline, "Best Practices in Hit-to-Lead Optimization," **2023**.
- AODI Chemical Science Lecture and Workshop, Nigeria, **2023**.
- Medicinal Chemistry Symposium, Pharmaceutical Society of Taiwan, **2022**.
- MOL2NET'21, Conference on Molecular, Biomedical, Computational and Network Science and Engineering, **2021**.
- 3rd Asian Pharmacometrics Network Symposium, **2021**.

- International Conference on Data Science in Biology, **2020**.

PRESENTATIONS

- Poster presenter, Computational Medicinal Chemistry School, Novartis Institutes for Biomedical Research, Cambridge MA, **2024**.
- Poster presenter, 2nd Canadian Peptide and Protein Community Virtual Meeting, “Emerging Peptide and Protein Science,” **2024**.
- Poster presenter, LatinXChem Twitter Conference, “Repurposing Drugs to Find HIV-1 Protease Inhibitors: A Virtual Study,” **2020**.

WORKSHOPS AND WEBINARS

- Covalent Modulators in Drug Discovery, 4th ACS MEDI Virtual Webinar, **2025**.
- CAiSMD: Computational Applications in Secondary Metabolite Discovery, Technische Universität Dresden, **2022**.
- Drug Discovery from Huge Chemical Space, WuXi AppTec Research Service Division, **2021**.
- StarDrop and SeeSAR Drug Discovery Workshop Series, Zastr Innovations, **2020**.
- Drug Discovery: New Leads from Old Drugs, Ramaiah University of Applied Sciences, **2020**.
- Computational Approach for Design of COVID-19 Inhibitors, TIFAR Riau, **2020**.
- Drug Discovery from Huge Chemical Space, WuXi AppTec Research Service Division, **2021**.
- Docking, QSAR and Molecular Dynamics, Ramaiah University of Applied Sciences with IEEE-EMB MSRIT and SRIGEN, **2020**.
- Strategies for Writing a Good Research Article, Manipal Academy of Higher Education with Indian Bureau of Administrators and Technocrats, **2020**.
- Recent Trends in Drug Discovery, Diagnostics and Therapeutics, VIGNAN’s Foundation for Science, Technology and Research with partners, **2020**.
- Short-Term Training Program: Hands-on Computational Drug Design, Karpaga Vinayaga College of Engineering and Technology, **2020**.
- International E-conference on Biotechnology, Bioinformatics and Biomedicine, AIMST University, **2020**.

PROFESSIONAL DEVELOPMENT/CERTIFICATIONS

- Drug Discovery, University of California San Diego ([Coursera](#)). Certificate, **2022**.
- Bloodborne Pathogen (BBP) Training for Research Labs, University of Michigan. Certificate of Achievement, **2023**.
- Chemical Laboratory Safety, University of Michigan. Certificate of Achievement, **2023**.
- Responsible Conduct of Research and Scholarship (RCRS), University of Michigan. Certificate of Achievement, **2023**.
- Laboratory Biosafety and Biosecurity, [Public Health Agency of Canada](#). Certificate of Participation, **2023**.
- Good Citation Behavior, Web of Science Academy ([Clarivate](#)). Certificate, **2023**.
- Effective Communication for Healthy Outcomes, YALI Network. Certificate, **2018**.

LEADERSHIP AND GOVERNANCE

- Founder and Executive Director, Raphael Onuku Foundation, **2024 to present**.
- Elected Representative, Biological and Health Sciences (Division 1), Rackham Student Government, University of Michigan, **2024 to 2025**.
- Student Advisory Board Member, Ginsberg Center, University of Michigan, **2024 to 2025**.
- President, Taipei Medical University International Students Association (TMU-ISA), **2022 to 2023**.

MEMBERSHIP AND PROFESSIONAL SERVICE

- Member, Sigma Xi, The Scientific Research Honor Society, **2026 to present.**
- Member, Alpha Chapter of the Rho Chi Society, **2026 to present.**
- Member, ACS Medicinal Chemistry Division, University of Michigan Student Chapter, **2023 to present.**
- Institutional membership, National Center for Faculty Development and Diversity (NCFDD), **2023 to present.**
- Student member, National Organization for the Professional Advancement of Black Chemists and Chemical Engineers (NOBCChE), **2023 to present.**
- Member, Visibility STEM Africa (VSA), **2020 to present.**
- Member, University of Nigeria Alumni Association, **2017 to present.**

SKILLS

- Computational chemistry: docking (Glide, Vina, MOE), MD (GROMACS, Desmond), modeling (MODELLER), visualization (PyMOL, VMD).
- Programming: Python automation; data analysis and reporting.
- Peptide chemistry: Fmoc SPPS; macrocyclization; HPLC and LC-MS.